ENTERPRISE AIOPS PLATFORM • ARCHITECTURE SPECIFICATION

Data Contracts, Canonical Schemas & Hybrid Cloud DLP

Target Platform: Google Cloud Platform (GCP) **System of Record:** ServiceNow ITSM **Version:** 2.0 (Production Blueprint) **Generated:** August 2026

Telemetry Data Contracts, Canonical Schemas & DLP Specifications

1. Overview & Objectives

In a heterogeneous enterprise observability ecosystem, each tool emits telemetry in proprietary, non-standard formats (e.g., Dynatrace PurePath JSON, Akamai DataStream records, Splunk CIM, Adobe HIT clickstream, GCP LogEntry).

To enable centralized ML anomaly detection, Vertex AI LLM correlation, and cross-domain root-cause analysis, the Ingestion Layer enforces a **Canonical Telemetry Event Contract**. All incoming raw streams are mapped and validated into this unified schema inside **Cloud Dataflow** before writing to **BigQuery** or publishing to the **Vertex AI Reasoning Bus**.

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2. Canonical Telemetry Event Schema

The Canonical Schema is formally defined in **Apache Avro** and **JSON Schema** formats, supporting backward and forward schema evolution.

2.1 JSON Schema Specification (`aiops.canonical.event.v1`)

```json
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "AIOpsCanonicalEvent",
  "type": "object",
  "required": ["event_id", "timestamp", "source_tool", "domain", "severity", "entity"],
  "properties": {
    "event_id": {
      "type": "string",
      "format": "uuid",
      "description": "Globally unique deterministic or UUIDv4 event identifier for deduplication."
    },
    "timestamp": {
      "type": "string",
      "format": "date-time",
      "description": "UTC ISO-8601 timestamp when the event occurred at source."
    },
    "ingestion_timestamp": {
      "type": "string",
      "format": "date-time",
      "description": "UTC ISO-8601 timestamp when GCP Dataflow processed the record."
    },
    "source_tool": {
      "type": "string",
      "enum": ["AKAMAI", "DYNATRACE", "GCP_OPS", "SPLUNK", "ADOBE_ANALYTICS"],
      "description": "The originating observability tool."
    },
    "domain": {
      "type": "string",
      "enum": ["EDGE_SECURITY", "APM_TRACES", "INFRASTRUCTURE", "ENTERPRISE_LOGS", "BUSINESS_TELEMETRY"],
      "description": "Functional observability domain."
    },
    "severity": {
      "type": "string",
      "enum": ["CRITICAL", "HIGH", "MEDIUM", "LOW", "INFORMATIONAL"],
      "description": "Normalized enterprise severity level."
    },
    "entity": {
      "type": "object",
      "required": ["service_name", "environment"],
      "properties": {
        "service_name": { "type": "string", "example": "checkout-api" },
        "environment": { "type": "string", "enum": ["production", "staging", "development"] },
        "host": { "type": "string", "example": "gke-prod-us-central1-pool-01" },
        "container_name": { "type": "string", "example": "checkout-service-pod-7d4f9" },
        "cloud_provider": { "type": "string", "enum": ["GCP", "AWS", "AZURE", "EDGE_AKAMAI", "ON_PREM"] },
        "region": { "type": "string", "example": "us-central1" },
        "cmdb_ci_id": { "type": "string", "description": "ServiceNow CMDB Configuration Item sys_id" }
      }
    },
    "metrics": {
      "type": "array",
      "items": {
        "type": "object",
        "required": ["name", "value", "unit"],
        "properties": {
          "name": { "type": "string", "example": "response_time_p99" },
          "value": { "type": "number", "example": 2450.5 },
          "unit": { "type": "string", "example": "ms" },
          "dimensions": { "type": "object", "additionalProperties": { "type": "string" } }
        }
      }
    },
    "log_payload": {
```

```
      "type": "object",
      "properties": {
        "message": { "type": "string", "description": "DLP-sanitized log message or exception string." },
        "stack_trace": { "type": "string", "description": "DLP-sanitized code call stack." },
        "trace_id": { "type": "string", "example": "4bf92f3577b34da6a3ce929d0e0e4736" },
        "span_id": { "type": "string", "example": "00f067aa0ba902b7" },
        "error_code": { "type": "string", "example": "HTTP_504_GATEWAY_TIMEOUT" }
      }
    },
    "business_context": {
      "type": "object",
      "properties": {
        "orders_per_minute": { "type": "number", "example": 420.0 },
        "cart_abandonment_rate": { "type": "number", "example": 0.185 },
        "estimated_revenue_loss_usd": { "type": "number", "example": 15400.0 },
        "funnel_stage": { "type": "string", "enum": ["BROWSE", "CART", "CHECKOUT", "PAYMENT"] },
        "user_cohort": { "type": "string", "example": "iOS_App_v18.2" }
      }
    },
    "raw_attributes": {
      "type": "object",
      "description": "Original raw tool attributes retained for forensics (DLP sanitized).",
      "additionalProperties": true
    }
  }
}
```

3. Tool-to-Canonical Field Mapping Matrix

Canonical Field	Akamai Mapping	Dynatrace Mapping	GCP Ops Suite Mapping	Splunk Mapping	Adobe Analytics Mapping
`event_id`	`reqId`	`problemId` / PurePath Span ID	`insertId`	`_cd` / `event_id`	`hitid_high` + `hitid_low`
`timestamp`	`start` (epoch)	`startTime`	`timestamp`	`_time` (epoch)	`date_time` (UTC)
`source_tool`	`"AKAMAI"`	`"DYNATRACE"`	`"GCP_OPS"`	`"SPLUNK"`	`"ADOBE_ANALYTICS"`
`domain`	`"EDGE_SECURITY"`	`"APM_TRACES"`	`"INFRASTRUCTURE"`	`"ENTERPRISE_LOGS"`	`"BUSINESS_TELEMETRY"`
`severity`	Calculated from WAF action / HTTP 5xx	Davis AI Severity (`AVAILABILITY`, `PERFORMANCE`)	`severity` (`CRITICAL`, `ERROR`)	`urgency` / `alert_level`	Anomaly deviation magnitude (Z-score)
`entity.service_name`	`cpCode` / Host header	Dynatrace `entityName` / Service Tag	`resource.labels.container_name`	`source` / `index`	`reportSuiteID` / App ID
`metrics`	TTFB, Bytes Transferred, WAF rule score	Response Time (p95/p99), CPU %, GC pause ms	Container CPU, Memory limit %, Pub/Sub lag	Transaction volume, Search runtime	Orders Per Minute (OPM), Cart Adds, GMV
`log_payload.trace_id`	`edgeTraceId`	PurePath `traceId`	`trace` (`projects/…/traces/…`)	CIM field `trace_id`	`marketing_cloud_visitor_id`
`business_context`	Geolocation, ISP, Bot Score	User Session tags, Cart Value	N/A	POS store ID, Terminal ID	Orders/Min, Cart Drop %, Funnel stage

4. Cloud DLP (Data Loss Prevention) Sanitization Rules

To ensure strict compliance with **PCI-DSS**, **GDPR**, and **CCPA**, all incoming text payloads, log messages, stack traces, and query strings pass through **Google Cloud Data Loss Prevention (Cloud DLP)** inside the Dataflow pipeline before persistent storage in BigQuery.

4.1 In-Flight DLP De-Identification Templates

4.2 Configured InfoTypes & Transformation Strategies

InfoType Category	Detected Patterns	Transformation Method	Output Format in BigQuery
Credit Card Numbers	Visa, MasterCard, Amex, Discover	**Masking**: Mask all characters except last 4	************1234
CVV / CVV2	3-4 digit card security codes	**Complete Redaction**	[CVV_REDACTED]
Email Addresses	User email headers and query strings	**Crypto-Deterministic Hash (HMAC-SHA256)**	hash_e4a8b71 ... (enables cross-session cohorting without PII exposure)
Phone Numbers	US/International phone formats	**Character Replacement**	[PHONE_REDACTED]
Authentication Tokens	Bearer tokens, API Keys, JWT signatures	**Regex Redaction**	Bearer [TOKEN_REDACTED]
Passwords / Passphrases	Form body passwords, connection strings	**String Replacement**	password=[REDACTED]

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5. Schema Evolution & Governance

1. **Forward Compatibility**: New optional fields can be added without breaking existing Dataflow stream parsers.
2. **Schema Registry**: Canonical Avro schemas are stored and versioned in a centralized GCP Cloud Storage schema registry bucket (gs://aiops-schema-registry/v1/).
3. **Validation Enforcement**: Messages failing canonical JSON Schema validation are rejected by Dataflow and directed to the Dead-Letter Queue (telemetry.<source>.dlq) with an attached validation_error_code .